Classification codes

Cooling down

Task code **4EC018**

Description This practical task requires students to investigate how quickly hot water cools down in different coloured containers. Students are then asked to consider other factors which may affect the cooling process.

Conceptual strand / context ☐ Natural and processed materials ☐ Life and living ☒ Energy and change ☐ Earth and beyond

Task type ☐ Objective ☐ Open ended ☒ Practical

Learning outcome focus ☐ Concept ☒ Process

Task focus level ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

Assessment purpose ☐ Diagnostic ☒ Formative ☒ Summative

Keywords temperature, time, conduction, radiation

Items	1	2	3	4	5	6
Highest scientific literacy level	2	3	3	4	3	3
Conceptual strand / context	EC	EC	EC	EC	EC	EC
Domain	B	B	B	C	C	A
Response type	Open ended	Open ended	Open ended	Open ended	Open ended	Open ended
Assessment purpose	Summative	Summative	Summative	Summative	Summative	Summative

Items	7	8	9	10
Highest scientific Literacy Level	4	4	4	3
Conceptual strand / context	EC	EC	EC	EC
Domain	B	B	B	A
Response type	Open ended	Open ended	Open ended	Open ended
Assessment purpose	Summative	Formative	Formative	Summative

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Equipment required

Per group:

- Three polystyrene cups (one painted black, one normal white, one with aluminium foil wrapped around it) with a 'fill to' level marked inside each one. Variations on this are possible: aluminium drink cans for example, and a measured amount of water rather than a 'fill to' level.
- Three thermometers.
- Measuring jug.
- Water supply–hot water (teacher to supply in a jug – maximum temperature 50^0C).
- Three retort stands, three bossheads, and three clamps if available.

Teacher-led instructions for practical task

1. Warn students about the dangers of working with hot water.
2. Remind students to let the thermometer settle before they note the reading. (Remind students that they must not use the thermometer as a stirring rod.)
3. This experiment is approximate only – it is not expected to be extremely accurate, as long as students get the idea.
4. The teacher could demonstrate the procedure for the whole class.

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Cooling down 4EC018

Task / Teacher record sheet

Student: _____

Collect the equipment from your teacher.

Look at the three thermometers. Write down the temperature shown by each thermometer.

Thermometer 1 _____ ^{0}C, Thermometer 2 _____ ^{0}C, Thermometer 3 _____ ^{0}C

4EC018Q1

Q1 What is the approximate room temperature?
_____ ^{0}C

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Set up three polystyrene cups – one normal, one painted black and one with aluminium foil wrapped around it.



Normal Black Aluminium foil

You or your teacher must put hot water into each of the three cups up to the fill mark.

Put a thermometer into each cup. Make sure it is stable – hold it at the tip, or use a boss head clamp and retort stand.

Measure the temperature of the water in each cup. Allow the thermometer a few seconds to settle at a temperature. (Do not use the thermometer as a stirring rod.)

4EC018Q2

Q2 Complete the row in the table for the starting time (0 minutes).
Every three minutes, repeat the measurements and record your results.

	Temperature ^{0}C		
Time (minutes)	**Normal**	**Black**	**Aluminium**
0			
3			
6			
9			
12			
15			

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4EC018Q3

Q3 Which colour cup cooled down the most?

_____

4EC018Q4

Q4 Why do you think the cup in **Q3** cooled down the most?

_____

_____

4EC018Q5

Q5 Was the first temperature measurement for the water in each cup the same?
If not, why not?

_____

_____

4EC018Q6

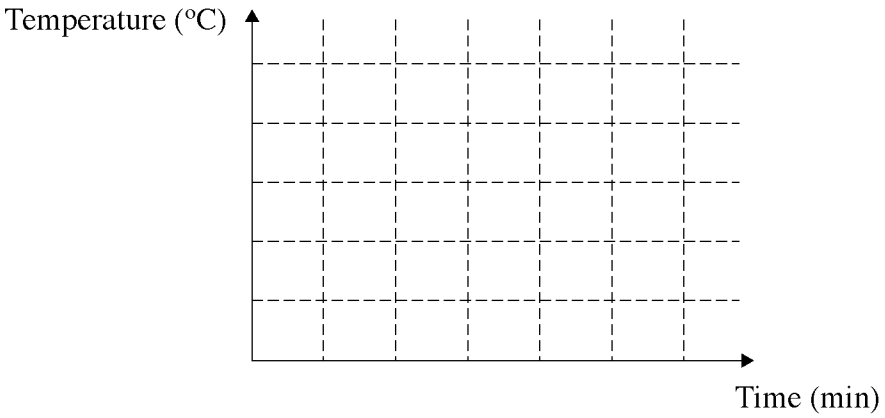
Q6 Why did you use the same amount of hot water in each cup?

_____

_____

4EC018Q7

Q7 Plot a line graph of temperature against time. Put all three sets of results on the same axes.



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4EC018Q8

Q8 What do the graphs show you?

_____

_____

4EC018Q9

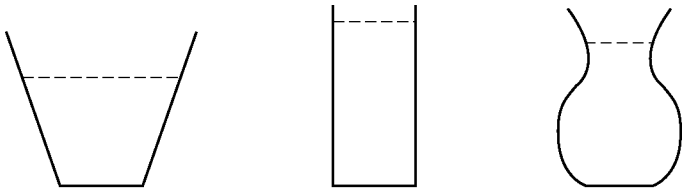
Q9 Imagine that the cups are left for **a few hours**. What do you think **the final temperature** of the water in each cup will be?

_____

_____

4EC018Q10

Q10 The three cups shown below are made from the same material. They have the same amount of hot water in them. Which one do you think will cool the quickest? Explain.



Cup A Cup B Cup C

_____

_____

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Marking guide

4EC018Q1

Q1 Score 1 *[2B–Makes comparisons between events observed.]*

Probably around 20^0C, but answers could vary between 10^0C and 30^0C (average of three thermometer readings if they are different).

Score 0 Other.

4EC018Q2

Q2 Score 1 *[3B–Displays data as a table.]*

Table fully completed.

Score 0 Other.

4EC018Q3

Q3 Score 1 *[3B–Identifies patterns in the scientific data.]*

Black.

Score 0 Other.

4EC018Q4

Q4 Score 1 *[4C–Explains process reported in terms of an abstract science concept.]*

Dark colours radiate heat quicker.

Score 0 Other.

4EC018Q5

Q5 Score 1 *[3C–Describes a cause and effect relationship.]*

The initial water temperatures should be the same, but they could vary (cool down) during the time it takes to pour the water into each cup.

Score 0 Other.

4EC018Q6

Q6 Score 1 *[3A–Demonstrates awareness of the need for fair scientific testing.]*

The same amount of water means it is a fair test.

Actual student responses:

- Because the more hot water there is the more heat will stay, thus meaning the experiment is unfair.
- Because we want to see how temperature changes through the appearance of cups, not through different volumes of water.

Score 0 Other.

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4EC018Q7

Q7 Score 1 *[4B–Plots line graphs.]*

Three lines on one graph with scale shown and correctly showing gradual decrease in temperature.

Score 0 Other.

4EC018Q8

Q8 Score 2 *[4B–Interprets data from line graph.]*

All three curves have a similar shape, but they cool at slightly different rates. They all show gradually decreasing temperature with time. They all approach the same final temperature.

Score 1 *[3B–Identifies patterns in science data.]*

All three curves have a similar shape and the temperatures are getting less.

Actual student response:

- The graphs show me the temperature dropped in each type of cup in every three minutes.

Score 0 Other.

4EC018Q9

Q9 Score 2 *[4B–Identifies patterns in science data.]*

They will all be at room temperature.

Score 1 *[3B–Identifies patterns in science data.]*

They will all be the same temperature.

Score 0 Other.

4EC018Q10

Q10 Score 1 *[3A–Identifies the variable to be changed.]*

Cup A because it has the largest surface area exposed to the air.

Score 0 Other.

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